

AI-NATIVE DEVELOPMENT · 2026

From Prompts to Loop Engineering

The workflow shift: prompt → context → harness → loop

The four-era arc of AI-native development — and why the unit of work you own has climbed all the way up to the iteration loop.

12 slides · first-person field notes

THE REFRAME

You're not getting better at prompting

The skill didn't improve. **It moved.**

Code generation got cheap, so **validation — not generation — is the bottleneck** (*CircleCI via InfoQ, Jun 2026*). That single inversion pushes the work up the stack.



You're not getting better at prompting. The level you work at is **moving up the stack.**

Prompt -> context -> rig -> loop: each step automates the craft below it and pushes your effort up a level.

From Prompts to Loop Engineering -- the four-era arc

THE THROUGH-LINE

One staircase, four eras

- **Prompt engineering** — the wording of one request
- **Context engineering** — what the model sees
- **Harness engineering** — everything *around* the model
- **Loop engineering** — the iteration *cycle* itself

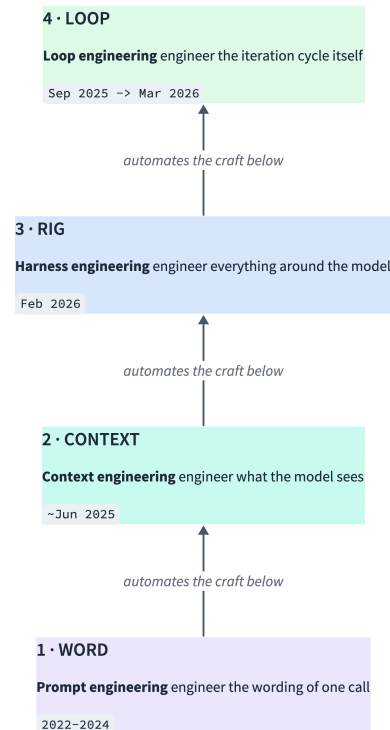
Each step automates the craft below it and pushes you up to govern a **bigger unit of work**.

By the end: know which step you're on, and ship your first real loop this week.

The four-era staircase

The staircase: four eras, one moving target

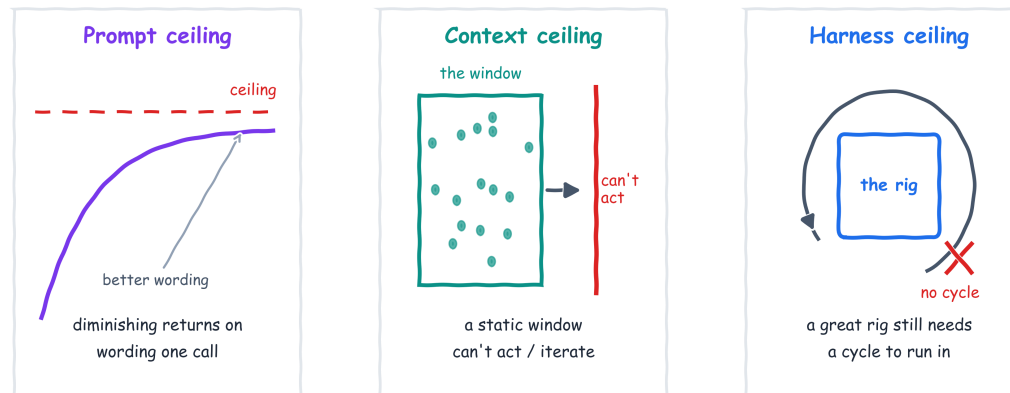
Each step automates the craft below it and pushes your effort up a level.



Every floor hits a ceiling

- **Prompt** — perfect phrasing, *still* a confident answer to the wrong problem
- **Context** — a well-fed model *still can't act*: it reads the repo, can't run the test
- **Harness** — mini-SWE-agent resolves **65% of tasks in ~100 lines** (*swebench.com, Jul 2025*)
— but a rig still needs a *cycle* to run in

Why each era hit a ceiling and pushed you up a level

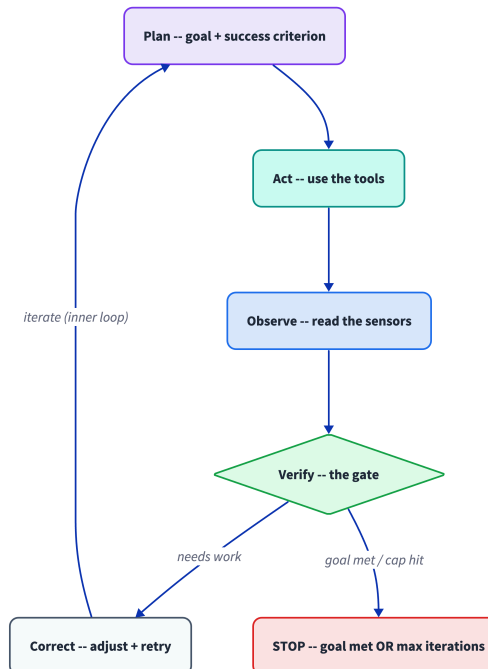


Proof tie-in: a minimal harness (mini-SWE-agent) hits 65% on SWE-bench Verified in ~100 lines (*swebench.com, Jul 2025*).

DEFINITION

What a loop actually is

Governing the agent's **iteration cycle** so it self-corrects *without a human in the inner loop*. Four levers: a **goal** with a success criterion · the **tools** to iterate · one **feedback signal** · a **stop condition** (VS Code “the agent loop”, May 2026; Anthropic, Dec 2024).

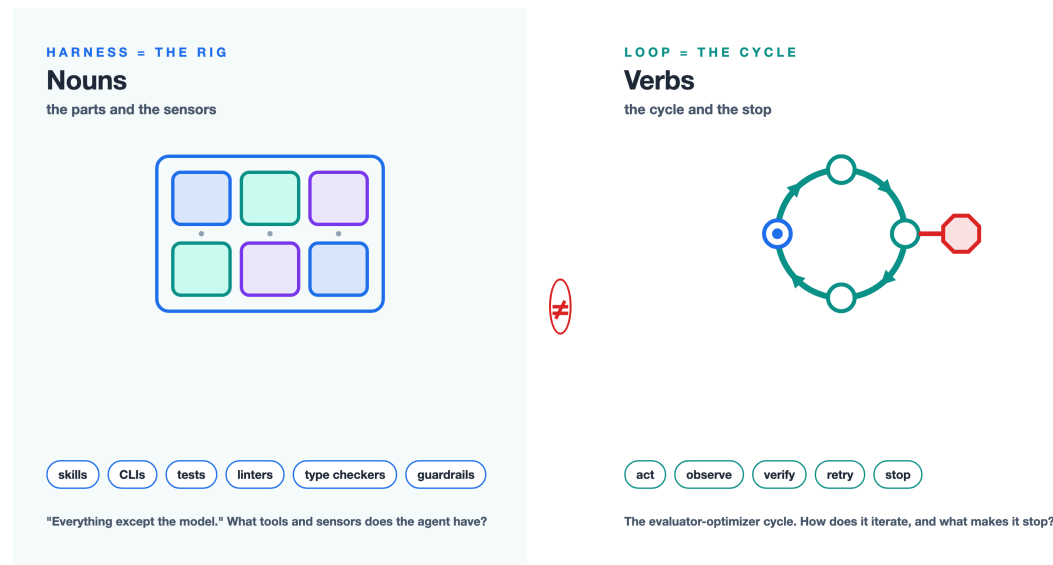


THE DISTINCTION EVERYONE BLURS

Harness vs. loop: nouns vs. verbs

- **Harness = the rig (nouns)** — context assembly + tool exposure + tool execution; tests, type checker, guardrails. VS Code: *"the harness is the product"*; Böckeler: *"everything except the model."*
- **Loop = the cycle (verbs)** — act → observe → verify → decide retry or stop.

The test: when you dislike the output, do you fix *the output* (editing) or change *the thing that produced it* (engineering)?



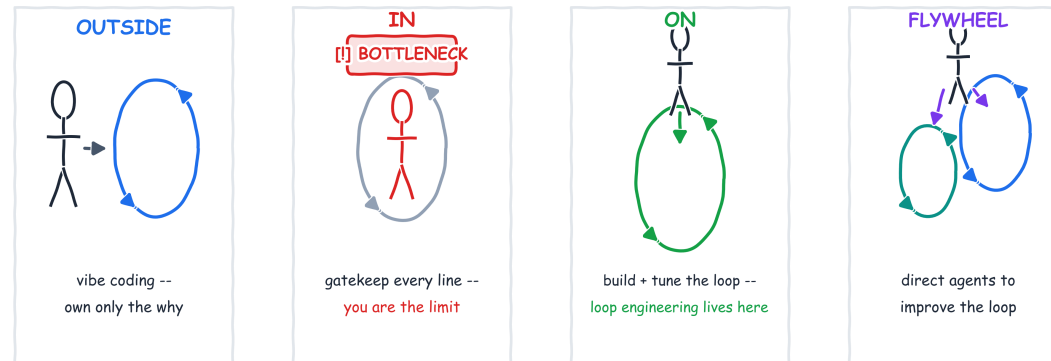
WHERE THE HUMAN SITS

Four postures around the loop

- **Outside** — vibe coding; you own only the *why*
- **In** — you gatekeep every line.
⚠️
The trap: you *become the bottleneck*
- **On** — you build and tune the cycle. *This is loop engineering.*
- **Flywheel** — you direct agents to improve the loop itself

(Kief Morris, "Humans and Agents in Software Engineering Loops", Mar 2026)

Who sits where: humans outside, in, on the loop -- and the flywheel

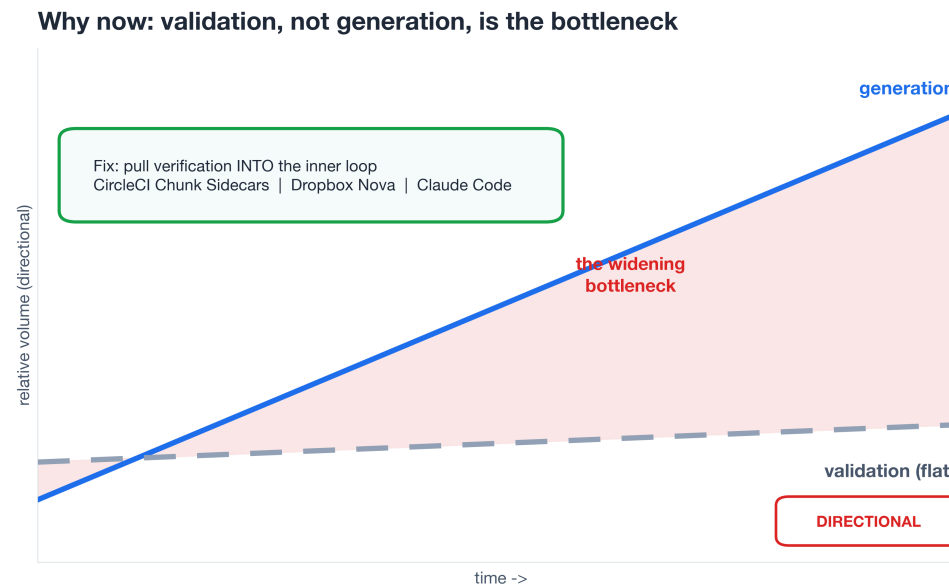


WHY NOW

Validation is the bottleneck

Generation throughput is rising; validation stayed flat. *"By the time conventional CI discovers an issue, the agent has already moved on, losing valuable context"* [\(CircleCI via InfoQ, Jun 2026\)](#).

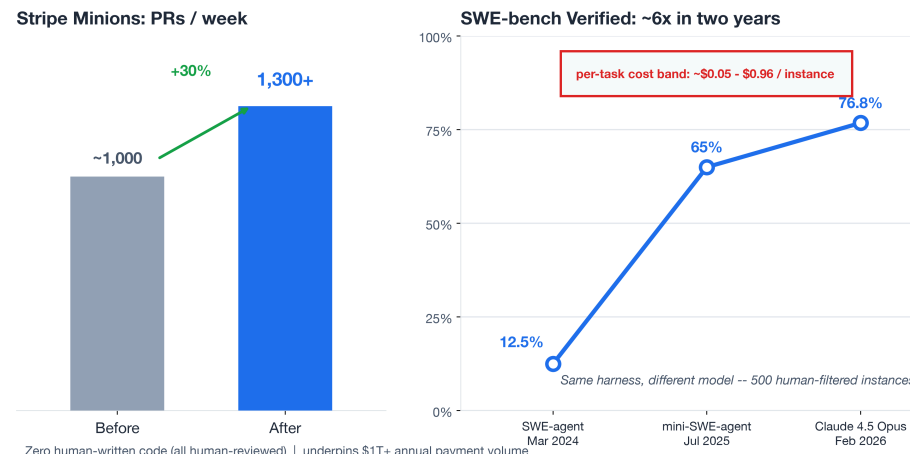
The fix: pull verification **into the inner loop** — CircleCI Chunk Sidecars, Dropbox Nova, Claude Code's iterative validation.



The industry numbers + loops you can read

- **Stripe: 1,300+ PRs/week, zero human-written code**, behind **\$1T+** in annual payment volume (*InfoQ* → *Stripe*, Mar 2026)
- **SWE-bench Verified, fixed harness: 12.47% → 76.8%** — illustrative; benchmark scores are harness-dependent (OpenAI stopped reporting it)
- **Read a real one: mini-swe-agent, Aider**, or **Azure/git-ape** (MIT) — open harness-plus-loops you can read end to end: plan → PR → deploy, with **security/cost gates** as sensors and CI/OIDC as the bounded run

Proof at scale: Stripe Minions and the SWE-bench trajectory



SHIP THIS WEEK

Your first agentic loop

Take one task you currently babysit line-by-line. Give it:

1. A clear **goal** with a success criterion
2. The **tools** to iterate — CLI, test runner, linter
3. One **machine-checkable feedback signal** — tests green, types clean
4. A **stop condition** — max iterations or a definition of done

Then step **on** the loop: next time it's wrong, *fix the loop, not the output.*

Your first loop: ship one this week

Take one task where you babysit the agent line-by-line, and give it four things.

1

A clear goal with a success criterion

2

The tools to iterate -- CLI, test runner, linter

3

One machine-checkable feedback signal

4

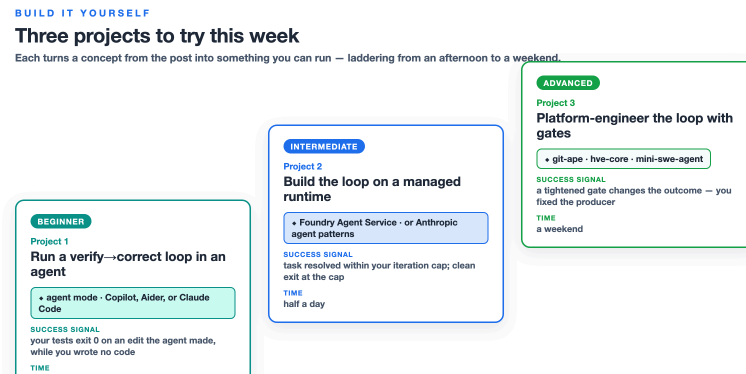
A stop condition -- max iterations or a definition of done

Source: Anthropic evaluator-optimizer + stop condition; this run's own loop.

BUILD IT YOURSELF

3 projects to try this week

- **Beginner — run a verify→correct loop in an agent** (Copilot, Aider, or Claude Code).
Success: your tests exit 0 on an agent-made edit. (*~90 min*) ·
`code.visualstudio.com/docs/agents`
- **Intermediate — build the loop on a managed runtime** (Foundry or Anthropic patterns).
Success: task resolved within your iteration cap; clean exit at the cap. (*half a day*) ·
`learn.microsoft.com` · Foundry Agent Service
- **Advanced — platform-engineer the loop with gates** (git-ape, hve-core, or mini-swe-agent):
tune a gate and re-run. Success: a tightened gate changes the outcome. (*a weekend*) ·
`github.com/Azure/git-ape`



TAKE THE NEXT STEP

Locate your step. Then climb.

Word, context, rig, or loop — the point where your effort matters keeps rising as models absorb more of the work.

Right now, that unit is the loop. Start with Project 1 this week.

Full four-era breakdown + the three build-it-yourself projects, with the Stripe + SWE-bench data and the practitioners who named the arc:

sendtoshailash.github.io/content-creation/blog/loop-engineering-ai-native-development.html